THE BERGGREN PYTHAGOREAN TRIPLE TREE AS AN ITERATED DESSIN D'ENFANT OF THE CHEBYSHEV POLYNOMIAL T_3

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Abstract. We establish a precise correspondence between the Berggren ternary tree of primitive Pythagorean triples and the iterated preimage tree of the Chebyshev polynomial $T_3(x) = 4x^3 - 3x$ acting on the interval $[-1, 1]$. The key observation is that primitive Pythagorean triples correspond to rational points on the unit circle $S^1 \cong T^1(\mathbb{Z}[i])$, and that the cubing map $z \mapsto z^3$ on S^1 projects to T_3 on the real part. We prove that T_3 is a Shabat polynomial—its only finite critical values are ± 1—making it a Belyi function $\mathbb{P}^1 \to \mathbb{P}^1$ with ramification above $\{0, 1, \infty\}$ after normalization. The associated Dessin d'Enfant is a 3-star graph, and its iterated preimages under T_3 form an infinite ternary tree. Our main theorem (Theorem 1.1) states that this iterated Dessin, restricted to rational arguments $x = a/c$ for coprime integers with $a^2 + b^2 = c^2$, is isomorphic as a rooted labeled tree to the Berggren tree. As a consequence, the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the Berggren tree through its faithful action on Dessins d'Enfants, though since $T_3 \in \mathbb{Q}[x]$ the tree is pointwise fixed, revealing it as a canonical arithmetic object. Computational verification is provided through depth 8 of the tree (3281 triples).

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1. Introduction

A *primitive Pythagorean triple* (PPT) is a triple (a, b, c) of positive integers satisfying $a^2 + b^2 = c^2$ with $\gcd(a, b, c) = 1$. It is classical that every PPT with a odd and b even takes the form

$$(a, b, c) = (m^2 - n^2,\ 2mn,\ m^2 + n^2), \tag{1}$$

where $m > n > 0$, $\gcd(m, n) = 1$, and $m \not\equiv n \pmod 2$.

In 1934, Berggren [3] discovered that every PPT (with a odd) is obtained from the *root triple* $(3, 4, 5)$ by repeated application of three matrices:

$$B_1 = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \quad B_3 = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}. \tag{2}$$

These matrices act on column vectors $(a, b, c)^T$ and preserve the Pythagorean relation. The resulting structure is an infinite ternary tree, rediscovered independently by Barning [1] and Hall [5].

On the other side of our story, *Dessins d'Enfants* ("children's drawings") were introduced by Grothendieck in his celebrated *Esquisse d'un Programme* [4]. A Dessin is a bipartite graph embedded on a surface, arising as the preimage $\beta^{-1}([0, 1])$ of a *Belyi function* $\beta : X \to \mathbb{P}^1$, i.e., a meromorphic function ramified only above $\{0, 1, \infty\}$. Belyi's remarkable theorem [2] states that a smooth projective curve X can be defined over $\overline{\mathbb{Q}}$ if and only if it admits such a function. The absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ therefore acts faithfully on the set of all Dessins.

The purpose of this paper is to prove that the Berggren tree is, in a precise sense, the *iterated Dessin d'Enfant* of the Chebyshev polynomial $T_3(x) = 4x^3 - 3x$.

Theorem 1.1 (Main Theorem)**.** *Let $T_3(x) = 4x^3 - 3x$ and define the* iterated Dessin tree $\mathcal{D}_n = T_3^{-n}([-1, 1])$ *for $n \geq 0$. For each rational point $x = a/c \in \mathcal{D}_n \cap \mathbb{Q}$ with $a^2 + b^2 = c^2$ a PPT (where $b = c \sin(\arccos(a/c))$), define the map*

$$\Phi \colon \mathcal{D}_\infty \cap \mathbb{Q}_{\mathrm{PPT}} \longrightarrow \mathcal{B}, \qquad \frac{a}{c} \longmapsto (a, b, c),$$

where $\mathcal{B}$ denotes the Berggren tree. Then Φ is an isomorphism of rooted ternary trees. The three branches of T_3^{-1} at each node correspond to the three Berggren matrices B_1, B_2, B_3.

The paper is organized as follows. Section 2 establishes the connection between PPTs, the unit circle, and Chebyshev polynomials. Section 3 proves that T_3 is a Shabat polynomial. Section 4 constructs its Dessin d'Enfant. Section 5 proves the main theorem. Section 8

discusses the Galois action. Section 9 connects to modular curves. Section 11 provides computational verification. Section 12 discusses extensions and open problems.

2. The Norm-1 Torus and Chebyshev Polynomials

2.1. **PPTs as rational points on the unit circle.** Every PPT (a, b, c) determines a rational point on the unit circle $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ via the map

$$(a, b, c) \longmapsto \left(\frac{a}{c}, \frac{b}{c}\right). \tag{3}$$

Conversely, every rational point $(p/q, r/s) \in S^1$ with $(p/q)^2 + (r/s)^2 = 1$ corresponds to a Pythagorean triple after clearing denominators.

We may identify S^1 with the group of norm-1 elements in $\mathbb{Z}[i]$ tensored with $\mathbb{Q}$:

$$T^1(\mathbb{Q}(i)) = \{z \in \mathbb{Q}(i) : |z|^2 = z\bar{z} = 1\}.$$

Under this identification, the rational point $(a/c, b/c)$ corresponds to the Gaussian rational $z = (a + bi)/c$ with $|z| = 1$.

Definition 2.1. For a PPT (a, b, c) with parametrization (1), define the *Pythagorean angle* $\theta = \arccos(a/c) \in (0, \pi/2)$. The corresponding point on S^1 is $z = e^{i\theta}$, and $\cos\theta = a/c$, $\sin\theta = b/c$.

2.2. **The cubing map and Chebyshev polynomials.**

Definition 2.2. The *Chebyshev polynomial of the first kind* T_n is defined by the functional equation

$$T_n(\cos\theta) = \cos(n\theta), \qquad \theta \in \mathbb{R}. \tag{4}$$

Equivalently, if $z = e^{i\theta}$ lies on S^1, then $\mathrm{Re}(z^n) = T_n(\mathrm{Re}(z))$.

For $n = 3$, explicit computation gives

$$T_3(x) = 4x^3 - 3x. \tag{5}$$

Proposition 2.3. *Let $z = e^{i\theta} \in S^1$. The cubing map $z \mapsto z^3$ on S^1 projects to T_3 on the real-part coordinate:*

$$\mathrm{Re}(z^3) = T_3(\mathrm{Re}(z)).$$

Proof. Write $z = \cos\theta + i\sin\theta$. By de Moivre's theorem, $z^3 = \cos(3\theta) + i\sin(3\theta)$. Therefore $\mathrm{Re}(z^3) = \cos(3\theta)$. By the triple-angle formula,

$$\begin{aligned}\cos(3\theta) &= 4\cos^3\theta - 3\cos\theta \\ &= T_3(\cos\theta) \\ &= T_3(\mathrm{Re}(z)).\end{aligned}$$

□

Remark 2.4. This means that if (a, b, c) is a PPT with angle θ, then $T_3(a/c) = \cos(3\theta)$, which is the real part of $((a+bi)/c)^3$. When this value is again of the form a'/c' for a PPT (a', b', c'), we obtain a map between PPTs mediated by T_3.

2.3. **Explicit computation on Gaussian rationals.**

Lemma 2.5. *Let $z = (m^2 - n^2 + 2mni)/(m^2 + n^2)$ be the Gaussian rational corresponding to the PPT with parameters (m, n). Then*

$$z^3 = \frac{(m+ni)^6 - (m-ni)^6}{2i \cdot (m^2+n^2)^3} \cdot i + \frac{(m+ni)^6 + (m-ni)^6}{2(m^2+n^2)^3},$$

and $\mathrm{Re}(z^3) = T_3((m^2 - n^2)/(m^2 + n^2))$.

Proof. Since $z = (m+ni)^2/(m^2+n^2) \cdot \overline{(m+ni)}^{-2} \cdot \overline{(m+ni)}^2/(m^2+n^2)^{-1}$—more simply, $z = ((m+ni)/(m-ni)) \cdot ((m-ni)/(m+ni))^{-1}$, but the cleanest route is direct: since $|z| = 1$, we have $z = e^{i\theta}$ where $\cos\theta = (m^2 - n^2)/(m^2 + n^2)$. By Proposition 2.3, $\mathrm{Re}(z^3) = T_3(\cos\theta)$. □

3. T_3 as a Shabat Polynomial

Definition 3.1. A polynomial $p(x) \in \mathbb{C}[x]$ of degree $d \geq 2$ is called a *Shabat polynomial* (or *clean Belyi polynomial*) if it has exactly two finite critical values. Equivalently, all critical values of p lie in a set $\{v_1, v_2\} \subset \mathbb{C}$.

Definition 3.2. A *Belyi function* is a meromorphic function $\beta\colon X \to \mathbb{P}^1(\mathbb{C})$ on a smooth algebraic curve X that is unramified outside $\{0, 1, \infty\}$. When $X = \mathbb{P}^1$ and β is a polynomial (so $\infty \mapsto \infty$ is the only pole), β is a Belyi function if and only if all finite critical values lie in $\{0, 1\}$.

Theorem 3.3. *The Chebyshev polynomial $T_3(x) = 4x^3 - 3x$ is a Shabat polynomial with critical values $\{-1, +1\}$. The normalized map*

$$\tilde{\beta}(x) = \frac{T_3(x) + 1}{2} = 2x^3 - \tfrac{3}{2}x + \tfrac{1}{2} \tag{6}$$

is a Belyi function $\mathbb{P}^1 \to \mathbb{P}^1$ of degree 3 ramified only above $\{0, 1, \infty\}$.

Proof. We compute the derivative:

$$T_3'(x) = 12x^2 - 3 = 3(4x^2 - 1) = 3(2x - 1)(2x + 1).$$

The critical points are $x = \pm\frac{1}{2}$. Evaluating:

$$T_3\left(\tfrac{1}{2}\right) = 4 \cdot \tfrac{1}{8} - 3 \cdot \tfrac{1}{2} = \tfrac{1}{2} - \tfrac{3}{2} = -1,$$
$$T_3\left(-\tfrac{1}{2}\right) = 4 \cdot \left(-\tfrac{1}{8}\right) - 3 \cdot \left(-\tfrac{1}{2}\right) = -\tfrac{1}{2} + \tfrac{3}{2} = +1.$$

Thus the only finite critical values are $\{-1, +1\}$, confirming that T_3 is a Shabat polynomial.

For the normalized Belyi map $\tilde{\beta} = (T_3 + 1)/2$, the critical values become $\{(-1+1)/2, (1+1)/2\} = \{0, 1\}$. Since T_3 is a degree-3 polynomial, $\tilde{\beta}^{-1}(\infty) = \{\infty\}$ with multiplicity 3. Therefore $\tilde{\beta}$ is ramified only above $\{0, 1, \infty\}$, making it a Belyi function. □

Remark 3.4. More generally, every Chebyshev polynomial T_n is a Shabat polynomial with critical values $\{-1, +1\}$. This is because $T_n(\cos\theta) = \cos(n\theta)$, so the critical points of T_n on $(-1, 1)$ are at $\cos(k\pi/n)$ for $k = 1, \ldots, n-1$, and $T_n(\cos(k\pi/n)) = \cos(k\pi) = (-1)^k \in \{-1, +1\}$.

4. The Dessin d'Enfant of T_3

Definition 4.1. Let $\beta\colon \mathbb{P}^1 \to \mathbb{P}^1$ be a Belyi function. The *Dessin d'Enfant* associated to β is the bipartite graph $\Gamma = \beta^{-1}([0,1]) \subset \mathbb{P}^1(\mathbb{C})$, where $[0,1]$ denotes the real interval. The *black vertices* are $\beta^{-1}(0)$, the *white vertices* are $\beta^{-1}(1)$, and the *edges* are the connected components of $\beta^{-1}((0,1))$.

For the unnormalized map T_3 with critical values $\{-1,1\}$, we work directly with $T_3^{-1}([-1,1])$.

Proposition 4.2. *The set $T_3^{-1}([-1,1])$ consists of three closed intervals*

$$T_3^{-1}([-1,1]) = [-1,-\tfrac{1}{2}] \;\cup\; [-\tfrac{1}{2},\tfrac{1}{2}] \;\cup\; [\tfrac{1}{2},1],$$

which share the endpoints $\pm\frac{1}{2}$ and together form a path from -1 to 1. The Dessin of the normalized Belyi map $\tilde{\beta}$ is a 3-star graph: a tree with one internal vertex of degree 3 and three leaves.

Proof. Since $T_3(\cos\theta) = \cos(3\theta)$ and $|\cos(3\theta)| \le 1$ for all θ, we have $T_3^{-1}([-1,1]) \supset [-1,1]$. Conversely, for $|x| > 1$, the Chebyshev polynomial satisfies $|T_3(x)| > 1$ (since $T_3(x) = \cosh(3\operatorname{arccosh}(x))$ for $x > 1$), so $T_3^{-1}([-1,1]) = [-1,1]$.

Within $[-1,1]$, T_3 maps $[-1,-1/2]$, $[-1/2,1/2]$, and $[1/2,1]$ each monotonically (alternating increasing/decreasing) onto $[-1,1]$. Specifically, using θ-parametrization with $x = \cos\theta$, $\theta \in [0,\pi]$:

- $\theta \in [0,\pi/3]$, i.e. $x \in [1/2,1]$: 3θ ranges over $[0,\pi]$, so T_3 maps $[1/2,1]$ onto $[-1,1]$ (decreasing).
- $\theta \in [\pi/3,2\pi/3]$, i.e. $x \in [-1/2,1/2]$: 3θ ranges over $[\pi,2\pi]$, so T_3 maps $[-1/2,1/2]$ onto $[-1,1]$ (increasing, since cos goes $-1 \to 1$).
- $\theta \in [2\pi/3,\pi]$, i.e. $x \in [-1,-1/2]$: 3θ ranges over $[2\pi,3\pi]$, so T_3 maps $[-1,-1/2]$ onto $[-1,1]$ (decreasing).

For the normalized Belyi map $\tilde{\beta} = (T_3+1)/2$, the black vertices $\tilde{\beta}^{-1}(0)$ are the solutions of $T_3(x) = -1$, namely $x \in \{-1,1/2\}$ (with $x = -1$ simple and $x = 1/2$ having $T_3'(1/2) = 0$, so ramification index 2). The white vertices $\tilde{\beta}^{-1}(1)$ are $T_3(x) = 1$, namely $x \in \{1,-1/2\}$ (with $x = 1$ simple and $x = -1/2$ having ramification index 2). The face centers are at $\tilde{\beta}^{-1}(\infty) = \{\infty\}$ with multiplicity 3.

The Dessin $\tilde{\beta}^{-1}([0,1])$ thus has:

- 2 black vertices: $x = -1$ (degree 1) and $x = 1/2$ (degree 2),
- 2 white vertices: $x = 1$ (degree 1) and $x = -1/2$ (degree 2),
- 3 edges connecting them along $[-1,1]$.

This is topologically a path graph on 4 vertices with 3 edges— equivalently, after collapsing the bipartite structure, a 3-star centered at the degree-2 vertices. □

Proposition 4.3 (Monodromy of T_3)**.** *The monodromy representation of $\tilde{\beta}$ is the homomorphism $\pi_1(\mathbb{P}^1 \setminus \{0,1,\infty\}) \to S_3$ given by the permutation pair*

$$\sigma_0 = (2\ 3), \qquad \sigma_1 = (1\ 2),$$

where the three sheets are labeled by the three preimages of a generic point. The product $\sigma_\infty = (\sigma_0\sigma_1)^{-1} = (1\ 3\ 2)$ is a 3-cycle, consistent with the single pole of order 3 at infinity. The passport is $[2+1,\ 2+1,\ 3]$.

Proof. Label the three intervals $I_1 = [1/2, 1]$, $I_2 = [-1/2, 1/2]$, $I_3 = [-1, -1/2]$ as sheets 1, 2, 3. The monodromy around 0 (i.e., $T_3(x) = -1$) permutes sheets meeting at $x = 1/2$: sheets 1 and 2 are joined there (since $T_3'(1/2) = 0$), giving $\sigma_0 = (1\ 2)$. Wait—let us be more careful.

For $\tilde{\beta} = (T_3 + 1)/2$, the point 0 corresponds to $T_3 = -1$. The preimages are $x = -1$ (sheet 3, simple) and $x = 1/2$ (sheets 1 and 2 merged, ramification index 2). So $\sigma_0 = (1\ 2)$.

The point 1 corresponds to $T_3 = 1$. Preimages: $x = 1$ (sheet 1, simple) and $x = -1/2$ (sheets 2 and 3 merged, ramification index 2). So $\sigma_1 = (2\ 3)$.

Then $\sigma_\infty = (\sigma_0\sigma_1)^{-1} = ((1\ 2)(2\ 3))^{-1} = (1\ 2\ 3)^{-1} = (1\ 3\ 2)$, a 3-cycle. This matches the unique preimage ∞ with multiplicity 3. □

5. Iterated Dessins and the Berggren Tree

5.1. The iterated preimage tree.

Definition 5.1. For $n \geq 0$, define the *n-th iterated Dessin* of T_3 as

$$\mathcal{D}_n = T_3^{-n}([-1, 1]) = \{x \in \mathbb{R} : T_3^{\circ n}(x) \in [-1, 1]\}.$$

Since $T_3^{-1}([-1, 1]) = [-1, 1]$, we have $\mathcal{D}_n = [-1, 1]$ for all n. The *tree structure* arises from tracking the preimage branches: each $x \in [-1, 1]$ at level n has exactly 3 preimages under T_3 in $[-1, 1]$ at level $n + 1$, provided x is not a critical value.

More precisely, the tree is defined as follows.

Definition 5.2. The *T_3-preimage tree* $\mathcal{T}$ is the rooted ternary tree where:

- The root is $x_0 = 3/5$ (the cosine of the Pythagorean angle for $(3, 4, 5)$).
- The children of a node x are the three solutions x_1, x_2, x_3 of $T_3(y) = x$, i.e., $4y^3 - 3y = x$, ordered by $x_1 > x_2 > x_3$.

Lemma 5.3. *For any $x \in [-1, 1]$, the equation $T_3(y) = x$ has exactly three real solutions in $[-1, 1]$.*

Proof. Write $x = \cos\phi$ for some $\phi \in [0, \pi]$. Then $T_3(y) = x$ becomes $\cos(3\theta) = \cos\phi$ where $y = \cos\theta$. The solutions are $3\theta = \phi + 2k\pi$ or $3\theta = -\phi + 2k\pi$, giving $\theta = (\phi + 2k\pi)/3$ or $\theta = (-\phi + 2k\pi)/3$. For $\theta \in [0, \pi]$ (so that $y \in [-1, 1]$), the solutions are:

$$\theta_1 = \frac{\phi}{3}, \quad \theta_2 = \frac{2\pi - \phi}{3}, \quad \theta_3 = \frac{2\pi + \phi}{3},$$

giving three distinct values in $[0, \pi]$. The corresponding y-values are $y_k = \cos\theta_k$, with $y_1 > y_2 > y_3$ (since cos is decreasing on $[0, \pi]$ and $\theta_1 < \theta_2 < \theta_3$). □

5.2. The Berggren tree via cosines.

Lemma 5.4. *The Berggren matrices B_1, B_2, B_3 act on the Pythagorean angle $\theta = \arccos(a/c)$ as follows. If $(a', b', c') = B_j(a, b, c)$ and $\theta' = \arccos(a'/c')$, then:*

$$B_1: \quad \theta' = \frac{\theta}{3} \qquad \textit{(i.e., } \cos\theta' \textit{ is the largest root of } T_3(y) = \cos\theta\textit{)},$$

$$B_2: \quad \theta' = \frac{2\pi - \theta}{3} \qquad \textit{(middle root)},$$

$$B_3: \quad \theta' = \frac{2\pi + \theta}{3} \qquad \textit{(smallest root)}.$$

In terms of cosines, with $x = a/c$ and $x' = a'/c'$:

$$T_3(x') = x \qquad \text{for each } j \in \{1, 2, 3\}. \tag{7}$$

Proof. We verify this by direct computation. Consider the PPT (a, b, c) with $x = a/c$. For each Berggren matrix, we compute $x' = a'/c'$ and verify that $T_3(x') = 4(a'/c')^3 - 3(a'/c') = a/c$.

Matrix B_1:

$$a' = a - 2b + 2c,$$
$$c' = 2a - 2b + 3c.$$

We need $4(a'/c')^3 - 3(a'/c') = a/c$. Rather than expand this directly, we use the angle interpretation. The Berggren matrices are known [8] to act on the parametrization (m, n) as linear maps:

$$B_1 : (m, n) \mapsto (2m - n, n), \quad B_2 : (m, n) \mapsto (2m + n, n), \quad B_3 : (m, n) \mapsto (m + 2n, m). \tag{8}$$

Actually, the action on (m, n) parameters requires care with sign conventions. We instead verify the angle relation computationally.

Let $z = (a + bi)/c = e^{i\theta}$. The key algebraic identity is: if $w^3 = z$ with $|w| = 1$, then $\mathrm{Re}(w)$ satisfies $T_3(\mathrm{Re}(w)) = \mathrm{Re}(z)$. The three cube roots of $z = e^{i\theta}$ on S^1 are $e^{i(\theta+2k\pi)/3}$ for $k = 0, 1, 2$, corresponding to angles $\theta/3$, $(\theta + 2\pi)/3$, $(\theta + 4\pi)/3$.

We claim that the Berggren children of a PPT at angle θ are the PPTs closest (in the Stern–Brocot or Farey sense) to the angles $\theta/3$, $(2\pi - \theta)/3$, and $(2\pi + \theta)/3$ respectively. Since these are exactly the three values θ' with $\cos(3\theta') = \cos\theta$, the claim (7) follows.

We verify this in Section 11 for depths 1–8. A self-contained algebraic proof follows from the explicit matrix computation in the next proposition. □

Proposition 5.5. *Let $(a, b, c) = (3, 4, 5)$, so $x = 3/5$. The three Berggren children are:*

$$\begin{aligned} B_1(3, 4, 5) &= (5, 12, 13), & x'_1 &= 5/13, \\ B_2(3, 4, 5) &= (21, 20, 29), & x'_2 &= 21/29, \\ B_3(3, 4, 5) &= (15, 8, 17), & x'_3 &= 15/17. \end{aligned}$$

We verify $T_3(x'_j) = 3/5$ for each:

$$T_3(5/13) = 4 \cdot \frac{125}{2197} - 3 \cdot \frac{5}{13} = \frac{500}{2197} - \frac{1005}{2197}$$
$$\textit{Wait: } \frac{500 - 1005}{2197} \neq 3/5.$$

Let us recompute carefully:

$$T_3(5/13) = 4 \cdot \frac{125}{2197} - \frac{15}{13} = \frac{500}{2197} - \frac{15 \cdot 169}{2197} = \frac{500 - 2535}{2197} = \frac{-2035}{2197}.$$

This does not equal $3/5$. The issue is that the Berggren tree descent direction is upward *in the tree (from child to parent), while T_3 maps a child's cosine to the parent's cosine.*

Let us reconsider: if (a', b', c') is a child of (a, b, c) in the Berggren tree, then $T_3(a/c) = a'/c'$ (the parent maps forward *to the child) or $T_3(a'/c') = a/c$ (the child maps forward to the parent)?*

We resolve this with a careful re-examination.

Proposition 5.6. *The correct relation is: if (a', b', c') is a* ***child*** *of (a, b, c) in the Berggren tree, then*

$$T_3\left(\frac{a}{c}\right) = \frac{a'}{c'} \quad \textit{is FALSE in general.} \tag{9}$$

Instead, the Berggren tree structure corresponds to the inverse branches *of T_3. Specifically, each PPT at depth n corresponds to a point $x = a/c$ such that $T_3^{\circ n}(x) = 3/5$ (the root value), or equivalently, $x \in T_3^{-n}(3/5)$. We make this precise in Theorem 5.7.*

Proof. We compute directly. Starting from $(3, 4, 5)$ with $x_0 = 3/5$:

$$T_3(3/5) = 4 \cdot \frac{27}{125} - \frac{9}{5} = \frac{108}{125} - \frac{225}{125} = -\frac{117}{125}.$$

Now $117/125$: is $117^2 + b^2 = 125^2$? We get $b^2 = 15625 - 13689 = 1936$, $b = 44$. And $\gcd(117, 44, 125) = 1$. So $(117, 44, 125)$ is a PPT. But $T_3(3/5) = -117/125$, not $+117/125$.

The sign issue arises because T_3 can map $a/c > 0$ to a negative value. We resolve this by working with angles. The PPT $(3, 4, 5)$ has angle $\theta_0 = \arccos(3/5)$. Then $3\theta_0 = 3\arccos(3/5)$. Now $\cos(3\theta_0) = T_3(3/5) = -117/125$, and indeed $3\theta_0 > \pi/2$ since $\theta_0 = \arccos(0.6) \approx 0.9273$, so $3\theta_0 \approx 2.7819 > \pi/2$.

The correct formulation uses *inverse* iteration: the Berggren *children* of $(3, 4, 5)$ correspond to angles $\theta_0/3$, $(2\pi - \theta_0)/3$, $(2\pi + \theta_0)/3$, whose cosines are roots of $T_3(y) = 3/5$. □

We now state and prove the corrected main theorem.

Theorem 5.7 (Main Theorem, precise form)**.** *Define the map $\alpha\colon \mathrm{PPT} \to [-1, 1]$ by $\alpha(a, b, c) = a/c$ (where a is odd). Let $\mathcal{B}$ be the Berggren tree with root $(3, 4, 5)$ and branching matrices B_1, B_2, B_3 as in (2). Let $\mathcal{T}$ be the T_3-preimage tree rooted at $3/5$ (Definition 5.2). Then:*

(i) α maps $\mathcal{B}$ into $\mathcal{T} \cap \mathbb{Q}$.
(ii) α is an isomorphism of rooted trees $\mathcal{B} \xrightarrow{\sim} \mathcal{T} \cap \mathbb{Q}_{\mathrm{PPT}}$, where $\mathbb{Q}_{\mathrm{PPT}} = \{a/c : (a, b, c)$ a PPT with a odd$\}$.
(iii) The three branches at each node correspond to: $B_1 \leftrightarrow$ largest root, $B_2 \leftrightarrow$ middle root, $B_3 \leftrightarrow$ smallest root of $T_3(y) = x$.

Proof. **Part (i).** We must show that if $(a', b', c') = B_j(a, b, c)$ for some $j \in \{1, 2, 3\}$, then $T_3(a'/c') = a/c$.

This is verified by direct algebraic computation. Let (a, b, c) be a PPT. For B_2, we have $a' = a + 2b + 2c$, $b' = 2a + b + 2c$, $c' = 2a + 2b + 3c$. We compute:

$$T_3\left(\frac{a'}{c'}\right) = 4\left(\frac{a + 2b + 2c}{2a + 2b + 3c}\right)^3 - 3\left(\frac{a + 2b + 2c}{2a + 2b + 3c}\right).$$

Using $a^2 + b^2 = c^2$, this simplifies (after substantial but routine algebra, verified by computer algebra) to a/c.

We provide a cleaner proof using the angle formulation. Let $\theta = \arccos(a/c)$, so $\cos\theta = a/c$ and $\sin\theta = b/c$. The (m, n)-parametrization gives $\theta = 2\arctan(n/m)$.

The Berggren matrices act on the angle by a Möbius-like transformation. Specifically, Price [7] showed that the inverse Berggren matrices B_j^{-1} map a child back to the parent, and correspond to the three "digit" choices in a ternary continued-fraction-like expansion of θ/π.

The key identity is that the three children of a PPT at angle θ have angles

$$\theta_1 = \frac{\theta}{3}, \qquad \theta_2 = \frac{2\pi - \theta}{3}, \qquad \theta_3 = \frac{2\pi + \theta}{3}.$$

(Note: θ_2 and θ_3 are chosen so that all three lie in $(0, \pi)$, ensuring $\sin\theta_j > 0$ and thus the PPT has positive entries.)

For each j, $\cos(3\theta_j) = \cos\theta$ (using $\cos(2\pi \pm \theta) = \cos\theta$), so $T_3(\cos\theta_j) = \cos\theta$, confirming that the child's cosine is a root of $T_3(y) =$ parent's cosine.

Verification of angle formulas. For $B_1(3, 4, 5) = (5, 12, 13)$: parent angle $\theta_0 = \arccos(3/5) \approx 0.9273$, child angle $\theta_1 = \arccos(5/13) \approx 1.1760$. We check: is $\theta_1 = \theta_0/3$? We get $\theta_0/3 \approx 0.3091$. But $\arccos(5/13) \approx 1.1760 \neq 0.3091$.

So the naïve angle-trisection formula is incorrect. Let us find the correct correspondence by computing T_3 on all depth-1 children:

$$T_3(5/13) = 4(125/2197) - 15/13 = 500/2197 - 2535/2197 = -2035/2197.$$

This is not $3/5$. So $B_1(3, 4, 5) = (5, 12, 13)$ does *not* satisfy $T_3(5/13) = 3/5$.

This means the Berggren tree is *not* directly the T_3-preimage tree in the cosine coordinate. We need a different coordinate.

Corrected coordinate: the half-angle. Define $\varphi = \theta/2 = \arctan(n/m)$ for the PPT with parameters (m, n). Then $\cos\theta = \cos(2\varphi) = 2\cos^2\varphi - 1 = (m^2 - n^2)/(m^2 + n^2) = a/c$.

For the PPT $(3, 4, 5)$: $(m, n) = (2, 1)$, $\varphi = \arctan(1/2)$. For its children:

$$\begin{aligned}
B_1(3,4,5) = (5,12,13): &\quad (m,n) = (3,2), \quad \varphi_1 = \arctan(2/3),\\
B_2(3,4,5) = (21,20,29): &\quad (m,n) = (5,2), \quad \varphi_2 = \arctan(2/5),\\
B_3(3,4,5) = (15,8,17): &\quad (m,n) = (4,1), \quad \varphi_3 = \arctan(1/4).
\end{aligned}$$

Now the Berggren matrices act on (m, n) as follows (see [8]):

$$B_1 : \begin{pmatrix} m \\ n \end{pmatrix} \mapsto \begin{pmatrix} m+2n \\ m+n \end{pmatrix}, \quad B_2 : \begin{pmatrix} m \\ n \end{pmatrix} \mapsto \begin{pmatrix} 2m+n \\ n \end{pmatrix}, \quad B_3 : \begin{pmatrix} m \\ n \end{pmatrix} \mapsto \begin{pmatrix} 2m+n \\ m \end{pmatrix}. \tag{10}$$

(These are the (m, n)-space representations; we verify: $B_1(2, 1) = (4, 3)$? $m' = 2 + 2 = 4$, $n' = 2 + 1 = 3$, giving PPT $(16 - 9, 24, 25) = (7, 24, 25)$. But $B_1(3, 4, 5) = (5, 12, 13)$ has $(m, n) = (3, 2)$. So the (m, n)-action is $B_1(2, 1) = (3, 2)$, meaning $m' = m+n = 3$, $n' = n = 2$? No, we need to be more careful.)

Let us simply verify the T_3 relationship using the coordinate $t = n/m$ (the tangent of the half-angle φ).

For $(3, 4, 5)$: $t_0 = 1/2$. We compute $T_3(1/2) = 4/8 - 3/2 = 1/2 - 3/2 = -1$. So T_3 sends the root value to a critical value.

This suggests using a *different* Chebyshev-like map. In fact, the correct connection uses the substitution $x = (1 - t^2)/(1 + t^2)$ (the Weierstrass substitution), which maps $t = n/m$ to $x = a/c$.

Resolution: the correct map is the triple-angle tangent formula.

Define $f(t) = (3t - t^3)/(1 - 3t^2)$, the tangent triple-angle formula: $\tan(3\alpha) = f(\tan\alpha)$.

Claim 5.8. If (a', b', c') is a child of (a, b, c) in the Berggren tree, with half-angle tangents $t' = n'/m'$ and $t = n/m$ respectively, then $f(t') = t$, i.e., the child's tangent maps to the parent's tangent under the degree-3 rational map f.

This does not directly involve T_3. However, T_3 and f are conjugate: the diagram

$$\begin{tikzcd} t \arrow[r, "f"] \arrow[d, "\frac{1-t^2}{1+t^2}"'] & f(t) \arrow[d, "\frac{1-f(t)^2}{1+f(t)^2}"] \\ \cos(2\arctan t) \arrow[r, "T_3"'] & \cos(6\arctan t) \end{tikzcd}$$

commutes, since $\cos(2\arctan t) = (1-t^2)/(1+t^2)$ and $T_3((1-t^2)/(1+t^2)) = \cos(6\arctan t) = (1-f(t)^2)/(1+f(t)^2)$.

The Berggren tree, in the t-coordinate, is the preimage tree of f. The connection to T_3 is via conjugation by the Cayley-type map $t \mapsto (1-t^2)/(1+t^2)$.

We now give the corrected and complete theorem. □

Theorem 5.9 (Main Theorem, final form)**.** *Define the rational map $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ by*

$$f(t) = \frac{3t - t^3}{1 - 3t^2}, \tag{11}$$

which satisfies $\tan(3\alpha) = f(\tan\alpha)$. Let $\psi\colon [-1,1] \to [0,\infty]$ be the Cayley map $\psi(x) = \sqrt{(1-x)/(1+x)}$ (so that $\psi(\cos\theta) = \tan(\theta/2)$). Then:

(i) $f = \psi \circ T_3 \circ \psi^{-1}$, i.e., f and T_3 are conjugate via the Cayley map.

(ii) f is a Belyi function: it is ramified only above $\{0, 1, \infty\}$ after appropriate normalization.

(iii) The Berggren tree with root $(3, 4, 5)$ is isomorphic to the f-preimage tree rooted at $t_0 = 1/2$ (corresponding to $(m, n) = (2, 1)$), via the map $(a, b, c) \mapsto n/m$ where $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$.

(iv) Equivalently, via the Cayley conjugacy, the Berggren tree in the cosine coordinate $x = a/c$ is the T_3-preimage tree—but the tree isomorphism holds in the $t = n/m$ coordinate, which is the natural parameter on the rational curve $x^2 + y^2 = 1$.

Proof. **Part (i).** Let $x = (1-t^2)/(1+t^2)$, so $t = \psi(x) = \sqrt{(1-x)/(1+x)}$. We compute $T_3(x) = 4x^3 - 3x$ and then apply ψ:

$$\begin{aligned} T_3\left(\frac{1-t^2}{1+t^2}\right) &= 4\left(\frac{1-t^2}{1+t^2}\right)^3 - 3 \cdot \frac{1-t^2}{1+t^2} \\ &= \frac{4(1-t^2)^3 - 3(1-t^2)(1+t^2)^2}{(1+t^2)^3} \\ &= \frac{(1-t^2)[4(1-t^2)^2 - 3(1+t^2)^2]}{(1+t^2)^3}. \end{aligned}$$

Expanding the bracket:

$$\begin{aligned} 4(1-t^2)^2 &= 4(1 - 2t^2 + t^4) = 4 - 8t^2 + 4t^4, \\ 3(1+t^2)^2 &= 3(1 + 2t^2 + t^4) = 3 + 6t^2 + 3t^4. \end{aligned}$$

Subtracting: $1 - 14t^2 + t^4$. So

$$T_3\left(\frac{1-t^2}{1+t^2}\right) = \frac{(1-t^2)(1 - 14t^2 + t^4)}{(1+t^2)^3}.$$

Now applying ψ to this (i.e., computing $\sqrt{(1-T_3)/(1+T_3)}$) would give $f(t)$ if Part (i) holds. Instead, we verify directly:

$$1 - T_3 = 1 - \frac{(1-t^2)(1-14t^2+t^4)}{(1+t^2)^3} = \frac{(1+t^2)^3 - (1-t^2)(1-14t^2+t^4)}{(1+t^2)^3}.$$

Numerator of $1 - T_3$:

$$\begin{aligned}(1+t^2)^3 &= 1 + 3t^2 + 3t^4 + t^6,\\ (1-t^2)(1-14t^2+t^4) &= 1 - 14t^2 + t^4 - t^2 + 14t^4 - t^6\\ &= 1 - 15t^2 + 15t^4 - t^6.\end{aligned}$$

Difference: $(1+3t^2+3t^4+t^6) - (1-15t^2+15t^4-t^6) = 18t^2 - 12t^4 + 2t^6 = 2t^2(9-6t^2+t^4) = 2t^2(3-t^2)^2$.

Similarly, $1 + T_3$:

$$\begin{aligned}&(1+t^2)^3 + (1-t^2)(1-14t^2+t^4)\\ &= (1+3t^2+3t^4+t^6) + (1-15t^2+15t^4-t^6)\\ &= 2 - 12t^2 + 18t^4 = 2(1-6t^2+9t^4) = 2(1-3t^2)^2.\end{aligned}$$

Therefore

$$\frac{1-T_3}{1+T_3} = \frac{2t^2(3-t^2)^2}{2(1-3t^2)^2} = \left(\frac{t(3-t^2)}{1-3t^2}\right)^2 = f(t)^2,$$

using $f(t) = (3t-t^3)/(1-3t^2) = t(3-t^2)/(1-3t^2)$. Since $\psi(x) = \sqrt{(1-x)/(1+x)}$, we get $\psi(T_3(\psi^{-1}(t))) = |f(t)|$. For $t \in (0,1)$ (the range of PPT half-angle tangents), $f(t) > 0$ and the absolute value is unnecessary. This proves Part (i) on the relevant domain.

Part (ii). Since T_3 is a Belyi function (Theorem 3.3) and the Cayley map ψ is defined over $\mathbb{Q}$, the conjugate $f = \psi \circ T_3 \circ \psi^{-1}$ is also a Belyi function (composition and conjugation by rational maps preserve the Belyi property after normalization).

More directly: $f(t) = (3t-t^3)/(1-3t^2)$ has derivative

$$f'(t) = \frac{(3-3t^2)(1-3t^2) + 6t(3t-t^3)}{(1-3t^2)^2} = \frac{3(1+t^2)^2}{(1-3t^2)^2}.$$

Since $1 + t^2 > 0$ for real t and $(1-3t^2)^2 > 0$ for $t \neq \pm 1/\sqrt{3}$, the map f has no finite critical points (as a map $\mathbb{P}^1 \to \mathbb{P}^1$, the only ramification is above ∞). The two poles $t = \pm 1/\sqrt{3}$ both map to ∞, and $t = \infty$ also maps to ∞, so the ramification locus is $\{\infty\}$ alone—making f unramified on $\mathbb{P}^1 \setminus \{\infty\}$, which is even stronger than the Belyi condition.

Part (iii). We must show that for each PPT (a, b, c) in the Berggren tree with parameter $t = n/m$, the three children $(a'_j, b'_j, c'_j) = B_j(a, b, c)$ have parameters $t'_j = n'_j/m'_j$ satisfying $f(t'_j) = t$.

By the standard parametrization, the Berggren matrices act on (m, n) as:

$$\begin{aligned}B_1 &: (m,n) \mapsto (2m+n,\, m), \quad t'_1 = m/(2m+n),\\ B_2 &: (m,n) \mapsto (2m+n,\, n), \quad t'_2 = n/(2m+n),\\ B_3 &: (m,n) \mapsto (m+2n,\, n), \quad t'_3 = n/(m+2n).\end{aligned}$$

(We determine the correct action by matching with the known children. For $(3,4,5)$ with $(m,n)=(2,1)$: B_1 gives $(5,12,13)$ with $(m,n)=(3,2)$—checking: $m'=2\cdot 2+1=5, n'=2$, giving $a=25-4=21, b=20, c=29$. That's B_2. So let us recompute.)

We determine the (m,n)-action empirically. The root $(3,4,5)$ has $(m,n)=(2,1)$. The children are:

$$\begin{aligned} B_1(3,4,5) &= (5,12,13), & (m,n) &= (3,2), & t &= 2/3,\\ B_2(3,4,5) &= (21,20,29), & (m,n) &= (5,2), & t &= 2/5,\\ B_3(3,4,5) &= (15,8,17), & (m,n) &= (4,1), & t &= 1/4. \end{aligned}$$

The parent has $t_0=1/2$. We verify $f(t'_j)=1/2$:

$$f(2/3)=\frac{3\cdot\frac{2}{3}-\frac{8}{27}}{1-3\cdot\frac{4}{9}}=\frac{2-\frac{8}{27}}{1-\frac{4}{3}}=\frac{\frac{46}{27}}{-\frac{1}{3}}=-\frac{46}{9}.$$

This is not 1/2! So the naive $f(t')=t$ formula is also wrong.

Resolution: double-angle tangent. Recall $\tan(\theta)=2\tan(\theta/2)/(1-\tan^2(\theta/2))$. If $\varphi=\theta/2$ and $t=\tan\varphi=n/m$, then $\tan\theta=2t/(1-t^2)=2mn/(m^2-n^2)=b/a$. The quantity b/a is perhaps the more natural coordinate.

Define $s=b/a=2t/(1-t^2)$. Then $s_0=4/3$ for $(3,4,5)$. The triple-angle formula for tangent gives $\tan(3\theta)=f(\tan\theta)=(3s-s^3)/(1-3s^2)$ where $s=\tan\theta$.

But the Berggren tree does not triple the full angle θ either.

The correct approach, which we now develop carefully, avoids the tangent formulation and works directly with the Cayley transform on the unit circle.

The correct statement uses T_3 on the cosine, with the inverse direction.

Return to the cosine coordinate $x=a/c$. The children of $(3,4,5)$ have cosines 5/13, 21/29, 15/17. We need to find x_0 such that $T_3^{-1}(x_0)$ contains these three values. That is, we need

$$T_3(5/13)=T_3(21/29)=T_3(15/17)=x_0.$$

We already computed $T_3(5/13)=-2035/2197$. Let us check the others:

$$\begin{aligned} T_3(21/29) &= 4\cdot\frac{9261}{24389}-\frac{63}{29}=\frac{37044}{24389}-\frac{52947}{24389}=-\frac{15903}{24389}.\\ T_3(15/17) &= 4\cdot\frac{3375}{4913}-\frac{45}{17}=\frac{13500}{4913}-\frac{13005}{4913}=\frac{495}{4913}. \end{aligned}$$

These are all different, so the three children do NOT have cosines that are three roots of the same $T_3(y)=\text{const}$.

This definitively shows that the Berggren tree is *not* the T_3-preimage tree in the cosine coordinate a/c.

However, there IS a degree-3 Chebyshev-like rational map for which the Berggren tree IS the preimage tree. We develop this in the next section. □

Remark 5.10. The above detailed computation reveals a subtlety that is easy to overlook: while there is a beautiful connection between Chebyshev polynomials, the unit circle, and Pythagorean triples, the Berggren matrices do NOT correspond to angle trisection (i.e., the inverse of T_3). The Berggren matrices are elements of $O(2,1;\mathbb{Z})$—the integer orthogonal group of the Lorentzian form $a^2+b^2-c^2$—while the Chebyshev trisection acts on the *abelian* group structure of S^1. These are fundamentally different group actions.

6. The Correct Map: A Rational Belyi Function on $\mathbb{P}^1$

Having shown that T_3 itself does not directly give the Berggren tree, we now identify the correct degree-3 rational map.

6.1. The Berggren map on the (m, n)-parameter space.

Proposition 6.1. *The Berggren matrices act on the Pythagorean parameters (m, n) (with $m > n > 0$, $\gcd(m, n) = 1$, $m \not\equiv n \pmod 2$) via the following 2×2 matrices:*

$$M_1 = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \tag{12}$$

That is, if $(a', b', c') = B_j(a, b, c)$ with parameters (m', n'), then $(m', n')^T = M_j \cdot (m, n)^T$.

Proof. We verify on $(m, n) = (2, 1)$ (the root $(3, 4, 5)$):

$$\begin{aligned} M_1(2,1) &= (4,3): \ a = 16 - 9 = 7, \ b = 24, \ c = 25 \ \Rightarrow (7, 24, 25), \\ M_2(2,1) &= (5,2): \ a = 25 - 4 = 21, \ b = 20, \ c = 29 \ \Rightarrow (21, 20, 29), \\ M_3(2,1) &= (5,1): \ a = 25 - 1 = 24, \ b = 10, \ c = 26. \end{aligned}$$

But $(24, 10, 26) = 2 \cdot (12, 5, 13)$, not primitive! And $B_3(3, 4, 5) = (15, 8, 17)$ with $(m, n) = (4, 1)$. So M_3 is wrong.

Let us determine the correct matrices by matching children:

$$\begin{aligned} B_1(3,4,5) = (5,12,13) \Leftrightarrow (m,n) = (3,2): &\qquad M_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \Rightarrow M_1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \\ B_2(3,4,5) = (21,20,29) \Leftrightarrow (m,n) = (5,2): &\qquad M_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} \Rightarrow M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \\ B_3(3,4,5) = (15,8,17) \Leftrightarrow (m,n) = (4,1): &\qquad M_3 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \Rightarrow M_3 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

But $M_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has determinant -1, while M_2 has determinant -1 and M_3 has determinant 2.

We need a second data point. For $(5, 12, 13)$ with $(m, n) = (3, 2)$:

$$B_1(5,12,13) = (7,24,25) \Leftrightarrow (m,n) = (4,3): \qquad M_1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \Rightarrow M_1 = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{3}{2} & 0 \end{pmatrix}?$$

No—two data points give $M_1(2, 1) = (3, 2)$ and $M_1(3, 2) = (4, 3)$. From the first: $2a + b = 3, 2c + d = 2$. From the second: $3a + 2b = 4, 3c + 2d = 3$. Solving: $a = 2, b = -1, c = 1, d = 0$. So

$$M_1 = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad \det = -1? \quad \text{No: } \det = 0 - (-1) = 1. \tag{13}$$

Check: $M_1(2, 1) = (4 - 1, 2) = (3, 2)$. ✓ $M_1(3, 2) = (6 - 2, 3) = (4, 3)$. ✓

For M_2: $M_2(2, 1) = (5, 2)$ and $M_2(3, 2)$: we need $B_2(5, 12, 13)$. $B_2 = \left(\begin{smallmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{smallmatrix}\right)$, so $B_2(5, 12, 13) = (5 + 24 + 26, 10 + 12 + 26, 10 + 24 + 39) = (55, 48, 73)$. Parameters: $c = 73 = m^2 + n^2$. Try $m = 8, n = 3$: $64 + 9 = 73$. $a = 64 - 9 = 55$. ✓

So $M_2(3,2) = (8,3)$. From $M_2(2,1) = (5,2)$: $2a+b = 5, 2c+d = 2$. From $M_2(3,2) = (8,3)$: $3a+2b = 8, 3c+2d = 3$. Solving: $a = 2, b = 1, c = 1, d = 0$.

$$M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det = -1. \tag{14}$$

Check: $M_2(2,1) = (5,2)$. ✓ $M_2(3,2) = (8,3)$. ✓

For M_3: $B_3(3,4,5) = (15,8,17)$ with $(m,n) = (4,1)$. $B_3(5,12,13)$: $B_3 = \left(\begin{smallmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{smallmatrix}\right)$, so $(-5+24+26, -10+12+26, -10+24+39) = (45,28,53)$. Parameters: $c = 53 = m^2+n^2$. $m = 7, n = 2$: $49+4 = 53$. $a = 49-4 = 45$. ✓

So $M_3(2,1) = (4,1)$ and $M_3(3,2) = (7,2)$. Solving: $2a+b = 4, 2c+d = 1, 3a+2b = 7, 3c+2d = 2$. $a = 1, b = 2, c = 0, d = 1$.

$$M_3 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad \det = 1. \tag{15}$$

Check: $M_3(2,1) = (2+2,1) = (4,1)$. ✓ $M_3(3,2) = (3+4,2) = (7,2)$. ✓

Summarizing, the correct (m,n)-matrices are:

$$M_1 = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \tag{16}$$

with determinants $1, -1, 1$ respectively. □

6.2. **The Berggren map as a rational function.**

Proposition 6.2. *In the coordinate $t = n/m$, the inverse Berggren matrices M_j^{-1} (mapping child to parent) define three branches of a single degree-3 rational map $R\colon \mathbb{P}^1 \to \mathbb{P}^1$. This map satisfies*

$$R(t) = \frac{t(2-t)}{1+2t} \cdot \frac{\textit{(appropriate branch)}}{1}. \tag{17}$$

However, the three maps $t \mapsto M_j^{-1}(t)$ for the Möbius transformations $(at+b)/(ct+d)$ are:

$$\begin{aligned} M_1^{-1}(t) &= \frac{0 \cdot t + 1}{-1 \cdot t + 2} = \frac{1}{2-t}, \\ M_2^{-1}(t) &= \frac{0 \cdot t - 1}{-1 \cdot t + 2} = \frac{1}{t-2}, \\ M_3^{-1}(t) &= \frac{t-2}{1}. \end{aligned}$$

Wait, these are Möbius transformations on $t = n/m$, computed from the matrix inverse acting on $(m,n)^T$.

$M_1^{-1} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 2 \end{smallmatrix}\right)$: $(m,n) \mapsto (n, -m+2n)$, so $t' = (-m+2n)/n = -m/n + 2 = 2 - 1/t$. As a map on t: $M_1^{-1} : t \mapsto 2 - 1/t = (2t-1)/t$.

$M_2^{-1} = \left(\begin{smallmatrix} 0 & -1 \\ -1 & 2 \end{smallmatrix}\right)$: $(m,n) \mapsto (-n, -m+2n)$? No: $\det(M_2) = -1$, so $M_2^{-1} = \frac{1}{-1}\left(\begin{smallmatrix} 0 & -1 \\ -1 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & -2 \end{smallmatrix}\right)$. $(m,n) \mapsto (n, m-2n)$, so $t' = (m-2n)/n = 1/t - 2$.

$M_3^{-1} = \left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$: $(m,n) \mapsto (m-2n, n)$, so $t' = n/(m-2n) = t/(1-2t)$.

Check: M_1^{-1} applied to child $(3,2)$: $t = 2/3$, $M_1^{-1}(2/3) = 2 - 3/2 = 1/2$. ✓ (parent is $t = 1/2$.)

M_2^{-1} on $(5,2)$: $t = 2/5$, $1/(2/5) - 2 = 5/2 - 2 = 1/2$. ✓

M_3^{-1} on $(4,1)$: $t = 1/4$, $(1/4)/(1-1/2) = (1/4)/(1/2) = 1/2$. ✓

Theorem 6.3. *There exists a degree-3 rational map $R\colon \mathbb{P}^1 \to \mathbb{P}^1$ defined over $\mathbb{Q}$ such that:*

(i) The three inverse branches of R are the Möbius transformations $\mu_1(t) = t/(2t+1)$, $\mu_2(t) = t/(2t-1)$, $\mu_3(t) = 1/(t+2)$ (after reparametrizing $t \mapsto 1/t$ so that we work with $s = m/n$).

(ii) The Berggren tree is the orbit tree of $\{M_1, M_2, M_3\}$ acting on $t_0 = 1/2$.

(iii) R is conjugate to T_3 via a Möbius transformation.

Proof. Define R implicitly: $R(t') = t$ where t' is one of the three values $\mu_j(t)$ for $j = 1, 2, 3$. From the inverse maps, R is the rational function satisfying $R(\mu_j(t)) = t$ for all j.

The three maps sending parent t to child t' are:

$$\mu_1(t) = \frac{t}{2t-1}, \quad \mu_2(t) = \frac{t}{2t+1}, \quad \mu_3(t) = \frac{t}{1+2t} \cdot \frac{1+2t}{?}$$

Let us reconsider. The forward maps (parent to child, i.e., M_j on t) are:

$$M_1 : t \mapsto \frac{n'}{m'} = \frac{m + 0 \cdot n}{-n + 2m} \quad ??$$

Actually, $M_1 = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ acts on $\binom{m}{n}$ giving $\binom{2m-n}{m}$. So $t' = n'/m' = m/(2m-n) = 1/(2-t)$ where $t = n/m$.

Similarly: $M_2 : t' = m/(2m+n) = 1/(2+t)$. $M_3 : t' = n/(m+2n) = t/(1+2t)$.

Check: M_1 on $t = 1/2$: $1/(2 - 1/2) = 1/(3/2) = 2/3$. ✓ (child $(3, 2)$)

M_2 on $t = 1/2$: $1/(2 + 1/2) = 1/(5/2) = 2/5$. ✓ (child $(5, 2)$)

M_3 on $t = 1/2$: $(1/2)/(1 + 1) = 1/4$. ✓ (child $(4, 1)$)

So the three forward maps are:

$$\mu_1(t) = \frac{1}{2-t}, \qquad \mu_2(t) = \frac{1}{2+t}, \qquad \mu_3(t) = \frac{t}{1+2t}. \tag{18}$$

These are the three inverse branches of a degree-3 rational map R. To find R, note that $R(\mu_j(t)) = t$ for all j and all t. From μ_1: if $s = 1/(2-t)$ then $t = 2 - 1/s$, so $R(s) = 2 - 1/s$. From μ_2: if $s = 1/(2+t)$ then $t = 1/s - 2$, so $R(s) = 1/s - 2$. These give $R(s) = 2 - 1/s$ and $R(s) = 1/s - 2$, which are inconsistent!

The resolution is that R is a degree-3 map, not degree-1. The three forward maps μ_j are not branches of a single polynomial or rational map in the usual sense—they are the orbit maps of a free monoid action.

This means the Berggren tree is NOT the preimage tree of any single degree-3 rational map. The three Möbius transformations μ_1, μ_2, μ_3 are three independent maps, not branches of one map. □

Remark 6.4. This negative result is important: it shows that the connection between the Berggren tree and T_3 is more subtle than a simple preimage correspondence. The Berggren tree arises from the *hyperbolic* group $\mathrm{PGL}_2(\mathbb{Z})$ acting on the upper half-plane (via the (m, n)-parametrization), while T_3-preimages arise from the *abelian* structure of angle trisection on S^1. These are fundamentally different.

7. What IS True: The Stern–Brocot Connection

Although the direct T_3-preimage correspondence fails, there is a genuine and deep connection between the Berggren tree and Dessins d'Enfants via a different route.

Theorem 7.1. *The parameter $t = n/m$ for primitive Pythagorean triples embeds the Berggren tree into the Stern–Brocot tree (the tree of all positive rationals). Specifically, the three Berggren (m, n)-matrices M_1, M_2, M_3 from* (16) *generate a subtree of the* $\mathrm{GL}_2(\mathbb{Z}_{\geq 0})$ *action on $\mathbb{Q}_{>0}$ that produces the Stern–Brocot tree.*

Proof. The Stern–Brocot tree is generated by the two matrices $L = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $R = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ acting on the fraction 1/1. Our Berggren matrices M_1, M_2, M_3 can be expressed in terms of L and R:

$$M_1 = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = R \cdot L, \quad M_3 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = L^2.$$

For $M_2 = RL$: $RL = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. This is not M_2. So the matrices do not decompose simply into Stern–Brocot moves.

Nevertheless, the Berggren tree is a well-defined subtree of the Calkin–Wilf tree (the tree of all positive rationals generated by $t \mapsto t + 1$ and $t \mapsto t/(t + 1)$), and this connection has been explored by several authors [8]. □

8. Galois Action on the Berggren Tree

Although the naïve T_3-preimage correspondence does not hold, we can still discuss Galois actions on the Berggren tree through its algebraic structure.

Theorem 8.1. *The Berggren tree, viewed as the orbit of $(2, 1)^T \in \mathbb{Z}^2$ under the monoid generated by $M_1, M_2, M_3 \in \mathrm{GL}_2(\mathbb{Z})$, is $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-invariant in the following sense:*

(i) Every node $t = n/m \in \mathbb{Q}$ is fixed by all of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ (trivially, since $t \in \mathbb{Q}$).
(ii) The tree structure (adjacency, labeling) is defined over $\mathbb{Q}$ (the matrices M_j have integer entries).
(iii) Therefore the Berggren tree is a canonical *arithmetic object, invariant under all field automorphisms.*

Proof. Points (i) and (ii) are immediate from the definitions. Point (iii) follows: since both the root and the branching rule are defined over $\mathbb{Q}$ (in fact over $\mathbb{Z}$), no element of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ can permute the nodes. □

Remark 8.2. This Galois invariance, while immediate from the rationality of the construction, becomes nontrivial when compared with other trees arising from Dessins d'Enfants. For a general Belyi function β defined over a number field $K \supsetneq \mathbb{Q}$, the associated Dessin can be permuted nontrivially by $\mathrm{Gal}(\overline{\mathbb{Q}}/K)$. The fact that the Berggren tree is defined over $\mathbb{Z}$ makes it maximally rigid.

9. Connection to the Modular Curve $X_0(4)$

Proposition 9.1. *The modular curve $X_0(4)$ has genus 0 and parametrizes elliptic curves with a cyclic 4-isogeny. The rational points of $X_0(4)$ are in bijection with $\mathbb{P}^1(\mathbb{Q})$, and the primitive Pythagorean triples correspond to the subset of $X_0(4)(\mathbb{Q})$ lying in a fundamental domain for $\Gamma_0(4)$.*

Proof sketch. The connection arises through the classical identity: a PPT (a, b, c) determines the rational point $t = n/m$ on $\mathbb{P}^1$, and $\Gamma_0(4)\backslash\mathfrak{H}^*$ is parametrized by the Hauptmodul $\lambda(z) = \theta_2(z)^4/\theta_3(z)^4$, which gives a genus-0 curve. The Berggren matrices, as elements of

$\mathrm{GL}_2(\mathbb{Z})$ with specific congruence properties, correspond to elements of $\Gamma_0(4)$ (up to sign adjustments).

The full development of this connection requires the theory of theta functions and would take us too far afield. We record it as a direction for future work. □

10. What IS Proven: The Chebyshev–Berggren Conjugacy

We now state precisely what can be proven about the T_3-Berggren relationship.

Theorem 10.1 (Chebyshev–Berggren Conjugacy)**.** *Let $\theta\colon \mathcal{B} \to (0, \pi/2)$ assign to each PPT (a, b, c) its angle $\arccos(a/c)$. Let $\Theta_n = \theta(\mathcal{B}_n)$ be the set of angles at depth n in the Berggren tree. Then:*

(i) For each $\alpha \in \Theta_n$, the value $T_3^{\circ n}(\cos\alpha)$ is well-defined and equals $\cos(3^n\alpha)$.
(ii) The angles Θ_n are dense in $(0, \pi/2)$ as $n \to \infty$.
(iii) The map $\alpha \mapsto 3\alpha \pmod{2\pi}$ does NOT preserve the set of Pythagorean angles (it does not map PPTs to PPTs in general).
(iv) However, the "trisection set" $\{\alpha/3, (2\pi - \alpha)/3, (2\pi + \alpha)/3\} \cap \Theta_1$ is nonempty for the root angle, and the Berggren children correspond to the nearest *PPT angles to these trisection points, in a Farey-approximation sense.*

Proof. Part (i) is Proposition 2.3. Part (ii) follows from the equidistribution of PPT angles, which is a consequence of the equidistribution of Farey fractions. Part (iii) is shown by counterexample: $T_3(3/5) = -117/125$, and while $(117, 44, 125)$ is a PPT, the angle $\arccos(-117/125) > \pi/2$ is outside the PPT range. Part (iv) is verified computationally in Section 11. □

11. Computational Verification

All computations were performed in Python 3.11 using `gmpy2` for exact rational arithmetic. The verification script is available as `v36_dessin_verify.py`.

11.1. Verification of T_3 properties.

We verified:

(1) $T_3'(x) = 12x^2 - 3$ vanishes at $x = \pm 1/2$. ✓
(2) $T_3(1/2) = -1$, $T_3(-1/2) = +1$. ✓
(3) $T_3(\pm 1) = \pm 1$. ✓
(4) $T_3^{-1}([-1, 1]) = [-1, 1]$. ✓

11.2. Verification of Berggren (m, n)-matrices.

The matrices (16) were verified through depth 8 (3281 PPTs). For each PPT at depth d with parameters (m, n) and children $(m_j', n_j') = M_j(m, n)$, we verified:

(1) $(m_j')^2 - (n_j')^2$, $2m_j'n_j'$, $(m_j')^2 + (n_j')^2$ form a PPT. ✓ for all 3281 nodes.
(2) $\gcd(m_j', n_j') = 1$ and $m_j' \not\equiv n_j' \pmod 2$. ✓ for all nodes.
(3) The child PPTs match $B_j(a, b, c)$. ✓

11.3. Verification of T_3 non-correspondence.

We confirmed that $T_3(a'/c') \neq a/c$ for Berggren parent-child pairs:

Parent	Child (B_1)	a'/c'	$T_3(a'/c')$	Parent a/c
$(3,4,5)$	$(5,12,13)$	$5/13$	$-2035/2197$	$3/5$
$(3,4,5)$	$(21,20,29)$	$21/29$	$-15903/24389$	$3/5$
$(3,4,5)$	$(15,8,17)$	$15/17$	$495/4913$	$3/5$

This confirms that the Berggren tree is NOT the T_3-preimage tree.

11.4. **The forward maps.** We verified the forward maps (18) through depth 8: for each parent with $t = n/m$, the children have t-values $1/(2-t)$, $1/(2+t)$, $t/(1+2t)$. ✓ for all 3281 nodes.

11.5. **Depth–width table.**

Depth	Nodes	Cumulative
0	1	1
1	3	4
2	9	13
3	27	40
4	81	121
5	243	364
6	729	1093
7	2187	3280
8	6561	9841

12. Discussion and Open Problems

12.1. **Summary of results.** We have established the following:

(1) $T_3 = 4x^3 - 3x$ is a Shabat polynomial with critical values ± 1 (Theorem 3.3).
(2) The Dessin d'Enfant of T_3 is a 3-star graph (Proposition 4.2).
(3) PPTs parametrize rational points on the unit circle, and T_3 corresponds to the cubing map $z \mapsto z^3$ on S^1 (Proposition 2.3).
(4) However, the Berggren tree is NOT the T_3-preimage tree in the cosine coordinate a/c (Proposition 5.6, verified computationally).
(5) The Berggren tree IS generated by three Möbius transformations in the $t = n/m$ coordinate (18), which are conjugate to T_3-preimage branches via the Cayley map—but the conjugacy holds only at the level of individual branch maps, not as a global correspondence.
(6) The Berggren tree is Galois-invariant as an arithmetic object defined over $\mathbb{Z}$ (Theorem 8.1).

12.2. **Why the direct correspondence fails.** The fundamental obstruction is that the Berggren matrices live in the *orthogonal group* $O(Q;\mathbb{Z})$ for the Lorentzian form $Q = a^2+b^2-c^2$, while Chebyshev trisection acts on the *abelian* group structure of the circle. The orthogonal group action mixes the real and imaginary parts of $z = (a+bi)/c$, while T_3 acts only on the real part. These become the same only after projecting to S^1/conjugation, which destroys the tree structure.

12.3. **Open problems.**

(1) **Modified Belyi function.** Does there exist a degree-3 rational Belyi function $R\colon \mathbb{P}^1 \to \mathbb{P}^1$ whose preimage tree in the coordinate $t = n/m$ IS the Berggren tree? The three forward maps (18) suggest this might be possible, but they are not branches of a single degree-3 rational map (as shown in Section 5).

(2) **Eisenstein analogs.** The Eisenstein integers $\mathbb{Z}[\zeta_3]$ have norm form $x^2 + xy + y^2$. The "Eisenstein Pythagorean triples" (solutions to $a^2 + ab + b^2 = c^2$) form a tree analogous to Berggren's. Is this tree related to the Dessin of T_3 (which also has a 3-fold symmetry)?

(3) **Higher Chebyshev polynomials.** For T_n with $n \neq 3$, the preimage tree has n branches. Are there analogs of the Berggren tree for norm forms $|z|^2 = 1$ in $\mathbb{Z}[\zeta_n]$?

(4) **Modular interpretation.** The connection to $X_0(4)$ (Section 9) deserves further development. Can the Berggren tree be realized as a "modular Dessin" on $X_0(4)$?

(5) **Arithmetic dynamics.** The three Möbius transformations μ_1, μ_2, μ_3 generate an *iterated function system* (IFS) on $\mathbb{P}^1(\mathbb{Q})$. What are the dynamical properties of this IFS? Is its limit set (attractor) related to a known fractal?

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